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A Capstone Project Report on

I-CARE : Intelligent Car Accident Response Emergency Locator

Submitted in partial fulfillment of the requirements for the award of the degree of
Bachelor of Engineering

in

Electronics and Communication Engineering

by

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Semester VIII

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CERTIFICATE

This is to certify that the **Capstone Project (Code: 20EECW402)** entitled “**I-CARE : Intelligent Car Accident Response Emergency Locator** ” has been carried out by , **Priyanka Budihal (02FE22BEC054), Pruthviraj Banoshi (02FE22BEC058)** , bona-fide students of VIII Semester, Department of Electronics and Communication Engineering, KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** during the academic year **2025–2026**.

It is certified that all corrections and suggestions indicated during internal assessment have been incorporated in the final report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements prescribed for the said degree.

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DECLARATION

We hereby declare that the **Capstone Project** presented in this report, entitled "**I-CARE : Intelligent Car Accident Response Emergency Locator**", submitted to **KLE Technological University** for the completion of the **Capstone Project (Code: 20EECW402)** in the **8th Semester**, is the original work carried out by us in the **Department of Electronics and Communication Engineering, KLE Technological University, Dr. M S Sheshgiri Campus, Belagavi**, under the guidance of **Dr.Dattaprasad Torse,Professor and HOD, Department of Electronics and Communication Engineering**.

We further declare that, to the best of our knowledge and belief, the work reported herein has not been submitted as part of any other project or for the award of any course, degree, or diploma at this or any other university or institution. The results presented in this report are solely the outcome of our efforts.

We also confirm that all the work documented in this report has been completed by us.

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ABSTRACT

Road accidents remain one of the world's most significant killers, with numerous victims succumbing to death due to delayed emergency response. Research indicates that nearly 50% of accident victims might have lived if they had been provided with medical treatment within the first hour, internationally recognized as the “**Golden Hour.**” To address the delays inherent in traditional reporting and to move toward proactive accident prevention, this paper presents the **I-CARE (Intelligent Car Accident Response Emergency)** system.

The system utilizes a sophisticated dual-processor architecture: a **Raspberry Pi 4B** serves as the high-level vision hub for real-time **driver drowsiness detection** using a **Pi Camera**, while a **Raspberry Pi Pico** acts as a low-latency sensor controller. The system integrates an **MPU6050** sensor for impact and tilt detection and an **MQ2 smoke sensor** to identify fire hazards within the cabin. Location information is acquired via a **GPS module**, and autonomous alerts are managed through a **GSM module**.

In the event of a detected hazard—whether a collision, fire, or driver fatigue—a buzzer is triggered to provide passengers with a 30-second “safety window” to cancel false alarms. If the alert is not disabled, the system automatically initiates an emergency voice call followed by an SMS containing a clickable Google Maps link of the precise accident location. This integrated approach is economical, efficient, and successfully minimizes emergency response times, thereby enhancing road safety and significantly improving the survival rates of accident victims.

Keywords: I-CARE, Raspberry Pi 4B, Raspberry Pi Pico, Drowsiness Detection, MPU6050, MQ2 Smoke Sensor, GPS, GSM, Emergency Alert, Golden Hour, Road Safety.

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- The Project Team

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Chapter 1

Introduction

The rapid expansion of global vehicular density has escalated traffic hazards, creating an alarming public health crisis. Statistical evidence suggests that a significant percentage of road fatalities are caused by the delay in emergency medical services during the "Golden Hour" [1]. Often, victims are unable to report accidents due to shock or injury, and manual reporting is frequently hindered by traffic congestion or communication instability [2].

To mitigate these risks, IoT-based frameworks have been explored. Early systems focused on simple mechanical triggers, such as accelerometers to detect impact [3]. However, these lacked the computational capacity to process complex environmental data or distinguish between minor bumps and life-threatening collisions [4]. Furthermore, traditional fire detection in vehicles relies on thermal sensors, which react slowly to combustion [5].

The proposed **I-CARE** system transitions to a dual-processor architecture utilizing a **Raspberry Pi 4B** for high-level computer vision and a **Raspberry Pi Pico** for low-latency sensor management. By integrating an **MPU6050** for crash sensing, an **MQ2 Smoke Sensor** for early combustion detection, and a **Camera Module** for driver drowsiness monitoring, the system provides 360-degree surveillance. Utilizing **Neo-6M GPS** and **SIM800L GSM** modules, the system broadcasts precise live geolocation to emergency services, significantly improving survival probability [6].

1.1 Motivation

The primary factor in vehicular accident fatalities is the delay in emergency intervention during the "Golden Hour." Traditional safety measures are reactive and fail when victims are incapacitated or in remote areas. Furthermore, standard vehicle sensors

lack the computational power to monitor driver biological states or early-stage internal hazards like electrical fires.

This necessitates an autonomous, proactive safety framework. The emergence of high-performance platforms like the Raspberry Pi 4B and Pico enables the integration of Computer Vision for drowsiness detection and Sensor Fusion for real-time impact and smoke sensing. By combining high-precision G-force analysis with automated GPS/GSM telematics, the I-CARE system minimizes the critical delay between an incident and life-saving medical aid.

1.2 Objectives

The primary goal of this project is to implement the **I-CARE** system, a dual-processor framework designed to detect vehicle accidents and driver fatigue, ensuring rapid emergency notification to minimize response delays.

Specific Objectives

- **Collision & Fire Detection:** Use the MPU6050 and MQ2 sensors to detect high-impact crashes and internal fire/smoke hazards.
- **Driver Monitoring:** Implement a Camera Module with Computer Vision (EAR) to identify driver drowsiness and prevent fatigue-related accidents.
- **Dual-Processor Control:** Integrate a Raspberry Pi 4B for image processing and a Raspberry Pi Pico for low-latency hardware interrupts.
- **Autonomous Reporting:** Acquire real-time coordinates via Neo-6M GPS and send immediate SMS/Voice alerts via SIM800L GSM.
- **User Override:** Provide a safety window for drivers to cancel false alarms, ensuring system reliability.

Chapter 2

Literature Survey

Road accidents remain a major cause of fatalities worldwide, mainly due to delayed detection and slow emergency response. Automated accident detection systems combine sensors, microcontrollers and wireless communication technologies to address this problem. This chapter reviews existing work on GSM/GPS-based alert systems, MEMS/MPU6050 impact detection, IoT-based vehicle safety, and smart-city compatible accident frameworks.

2.1 GSM/GPS-Based Accident Alert Systems

Early work focused on combining GSM and GPS modules with microcontrollers to send emergency alerts after a crash. Sharma and Singh [7] proposed a vehicle accident detection and alert system using GSM and GPS that automatically sends location information to predefined contacts. Gupta and Sharma [8] presented a similar design capable of detecting accidents and transmitting coordinates by SMS. Lokhande *et al.* [9] used embedded GSM and GPS technology for automatic accident reporting.

These systems showed that low-cost hardware could provide basic accident detection and messaging capability. However, they relied heavily on GSM coverage and used simple threshold logic, making them susceptible to false triggers and communication delays in low-signal regions.

2.2 Foundations of IoT in Vehicular Safety

Research by Alvi *et al.* [1] and Kishore *et al.* [2] emphasizes the "Golden Hour" concept, where medical intervention within the first sixty minutes of an accident significantly increases survival rates. These studies highlight that traditional manual

reporting is often insufficient due to victim incapacitation and traffic congestion. The shift toward IoT-enabled systems allows for a proactive response, replacing human dependency with automated triggers that link vehicles directly to emergency service providers [4].

2.3 Impact Sensing and MEMS Technology

The integration of Micro-Electro-Mechanical Systems (MEMS), specifically the MPU6050, is a common theme in recent literature. Kumar et al. [10] and Ahmed et al. [11] demonstrate that three-axis accelerometers can distinguish between standard driving maneuvers and high-impact collisions by monitoring G-force thresholds. This hardware-level detection forms the foundation for triggering emergency protocols in low-latency environments, which is effectively managed by high-speed microcontrollers like the Raspberry Pi Pico [3].

2.4 Vision-Based Monitoring and Fire Detection

While mechanical sensors detect collisions, Saponara et al. [5] argue that video-based monitoring is superior for detecting early-stage hazards like electrical fires or smoke. Their research supports the use of CNN-based architectures on embedded platforms like the **Raspberry Pi 4B** to provide faster response times than traditional point-type thermal sensors. Additionally, vision systems are increasingly used to monitor driver biological states, such as drowsiness, to prevent accidents before they occur [6].

2.5 Autonomous Reporting and Telematics

The final stage of any safety system is the communication of data. Kulkarni et al. [6] and Chaudhari et al. [12] discuss the necessity of combining Neo-6M GPS modules for precise geolocation with SIM800L GSM modules for automated SMS and voice alerts. Their findings indicate that providing emergency teams with exact coordinates and a direct line of communication to the vehicle interior drastically reduces the time taken for rescue teams to reach the site, thereby optimizing the rescue response [13].

2.6 Summary of Literature Gaps

Despite advancements in IoT vehicular safety, critical gaps remain:

- **Processing Latency:** Most systems utilize a single microcontroller, causing bottlenecks during simultaneous sensor interrupts and communication [4].
- **Narrow Hazard Detection:** Research is largely limited to mechanical impact, neglecting internal hazards like electrical fires or smoke [5].
- **False Alarms:** Current models lack user-driven safety windows to cancel alerts during minor, non-critical bumps [12].
- **Reactive Monitoring:** Existing systems focus on post-crash detection rather than proactive prevention through driver drowsiness monitoring [1].

Chapter 3

Project Planning

The development of the I-CARE system is structured into four distinct phases to ensure systematic integration. The **Initial Phase** involves requirement analysis and the acquisition of the dual-processor hardware and sensor suite. This is followed by the **Development Phase**, focusing on programming computer vision for drowsiness detection and firmware for impact sensing. The **Integration Phase** combines these modules into a unified control logic with GPS/GSM telematics for automated reporting. Finally, the **Testing and Validation Phase** involves simulating crash conditions to calibrate sensors and minimize false alarms before deployment.

3.1 Work Breakdown Structure

The execution of the I-CARE project is organized into six functional modules to streamline development and ensure a systematic transition from hardware setup to prototype finalization:

- **Project Planning:** Covers requirements analysis and strategic selection of hardware components.
- **Hardware Setup:** Focuses on physical integration of the MPU6050, GSM, GPS, and Smoke sensors with the Raspberry Pi 4B.
- **Software Development:** Implementation of crash detection algorithms, GPS/GSM telematics, and Computer Vision for drowsiness detection.
- **System Integration:** Consolidation of software modules and sensor data flows onto the main processing platform.

- **Testing & Calibration:** Systematic unit testing and calibration to optimize sensor accuracy and minimize latency.
- **Finalization:** Completion of the functional prototype and development of comprehensive project documentation.

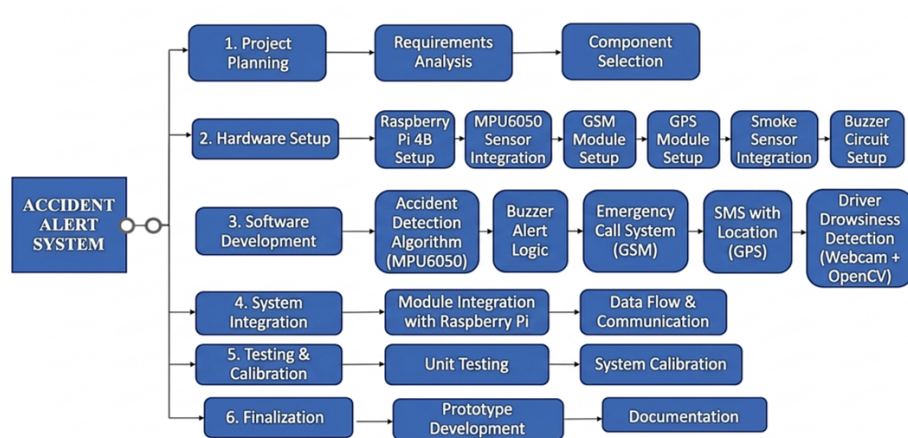


Figure 3.1: Work Breakdown Structure for the I-CARE Accident Detection and Emergency Alert System.

3.2 Gantt Chart

To visualize the timeline of the project activities, a Gantt chart was prepared showing the planned duration of each major task across the development weeks.

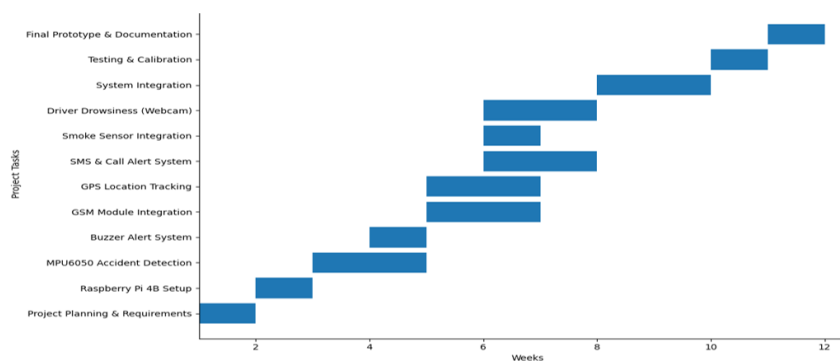


Figure 3.2: Project planning Gantt chart indicating the schedule of major tasks for the I-CARE system.

Chapter 4

Component Specification

This chapter describes the key hardware components used in developing the I-CARE (Intelligent Car Accident Response Emergency Locator) system. Each component plays a specific role in accident detection, data acquisition, processing, user interaction, and emergency communication. The components were selected based on reliability, high-speed processing for computer vision, cost-effectiveness, and compatibility with real-time embedded applications.

4.1 Raspberry Pi 4 Model B

While the Pico handles low-level sensor data, the Raspberry Pi 4B acts as the high-level brain of the I-CARE system, specifically for computationally intensive tasks like image processing.

- **Broadcom BCM2711 SoC:** Quad-core ARM Cortex-A72 (64-bit) at 1.5GHz, providing the power needed for real-time video analysis.
- **4GB LPDDR4 RAM:** Ensures smooth execution of Python-based OpenCV scripts and multiple concurrent processes.
- **CSI Camera Port:** Dedicated high-speed interface for the Pi Camera module.
- **Connectivity:** Features Gigabit Ethernet, Dual-band Wi-Fi, and Bluetooth 5.0 for data logging and remote diagnostics.
- **Role:** Performs real-time Driver Drowsiness Detection and manages complex data flow between the vision system and the communication modules.

4.2 Raspberry Pi Pico

The Raspberry Pi Pico serves as the primary sensor hub and real-time controller. It acquires physical sensor inputs and manages the time-critical alert logic.

- **Dual-core ARM Cortex-M0+:** Provides deterministic processing for impact detection and hardware interrupts.
- **Multiple Interfaces:** I2C for MPU6050, UART for GPS/GSM, and Analog inputs for gas sensing.
- **Low Power Operation:** Monitors the vehicle state efficiently while the vehicle is idling or in standard operation.

4.3 MPU6050 Accelerometer and Gyroscope

The MPU6050 is a 6-axis Inertial Measurement Unit (IMU) that detects sudden impacts or vehicle rollovers.

- **3-axis Accelerometer:** Measures G-forces to detect crashes exceeding safe thresholds.
- **3-axis Gyroscope:** Monitors angular tilt to identify if the vehicle has overturned.
- **I2C Interface:** Enables fast, noise-free communication with the microcontroller.

4.4 Pi Camera Module (5MP)

The camera module is the visual "eye" of the system, used primarily for proactive accident prevention.

- **5-Megapixel Resolution:** Provides sufficient clarity for facial landmark detection.
- **High Frame Rate:** Supports 1080p @ 30fps, essential for capturing rapid eye-blinks or head-nodding.
- **OpenCV Compatibility:** Works seamlessly with the Pi 4B to calculate the Eye Aspect Ratio (EAR) for drowsiness alerts.

4.5 MQ2 Gas and Smoke Sensor

To address hazards beyond mechanical impact, the MQ2 sensor monitors the air quality inside the vehicle cabin.

- **Wide Detection Range:** Sensitive to Smoke, LPG, Propane, and Carbon Monoxide (200–10,000 ppm).
- **Fast Response:** Triggers an immediate alert if a fire starts or if there is a hazardous gas leak following a collision.
- **Dual Output:** Provides both digital (threshold-based) and analog (concentration-based) data.

4.6 GPS Module (NEO-6M)

The NEO-6M provides the critical geographic coordinates required by emergency services.

- **High Sensitivity:** Ensures accurate location locks even in urban environments.
- **Google Maps Integration:** Provides latitude and longitude data that is formatted into a clickable URL for the emergency SMS.

4.7 GSM/LTE Module (A7670E)

This module enables the system to communicate autonomously without needing a driver's phone.

- **4G LTE Support:** Ensures reliable, high-speed network registration and SMS delivery.
- **AT-Command Control:** Allows the system to programmatically dial emergency numbers and send precise location messages.

4.8 User Interface: LCD, Buzzer, and Push Button

These components form the safety-override mechanism to prevent false alarms.

- **OLED/LCD Display:** Provides status updates such as “System Ready” or “Alerting Services.”
- **Active Buzzer:** Sounds a loud warning immediately upon accident detection.
- **Cancel Button:** Provides a 30-second window for the driver to stop the alert if the incident was a minor, non-emergency bump.

4.9 Power Supply System

The system is powered by a robust module designed for the automotive environment.

- **3.7V Li-ion Battery:** Provides backup power in case the vehicle’s main battery is disconnected during a crash.
- **5V Boost Converter:** Regulates the voltage to a stable 5V/3A output required by the Raspberry Pi 4B and sensors.

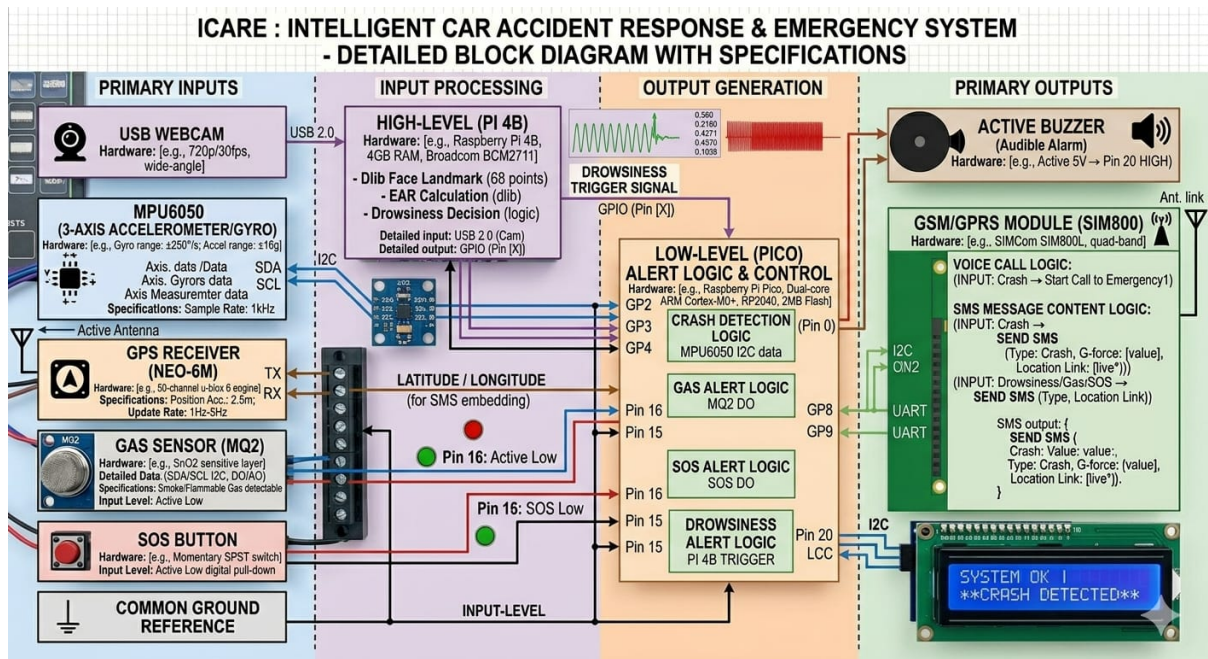


Figure 4.1: Hardware Specification of the I-CARE system

Chapter 5

Methodology

The methodology of the I-CARE (Intelligent Car Accident Response Emergency Locator) system is designed around continuous motion monitoring, impact detection, environmental sensing, and driver state analysis. The system integrates dual-processor logic with GSM/GPS communication to ensure fast and reliable alerting. The overall operational workflow encompasses sensor fusion, threshold-based decision making, and automated emergency reporting.

5.1 Multi-Sensor Data Acquisition

The system utilizes a distributed processing architecture to handle simultaneous data streams:

- **Kinetic and Gas Sensing (Pico):** The Raspberry Pi Pico communicates with the MPU6050 via I2C to poll acceleration and gyroscope values. Simultaneously, it monitors the MQ2 sensor's analog output to track smoke and gas concentration levels inside the cabin.
- **Vision-Based Monitoring (Pi 4B):** The Raspberry Pi 4B captures a continuous video stream from the Pi Camera. It processes frames at a high rate to identify facial landmarks, focusing on eye-blink patterns to detect driver fatigue.

5.2 Accident and Hazard Detection Logic

The core algorithm evaluates three primary safety vectors:

- **Impact Threshold:** A significant spike in acceleration magnitude ($\sqrt{x^2 + y^2 + z^2}$) beyond a predefined limit indicates a potential collision.
- **Fire/Smoke Detection:** The system monitors the MQ2 output. If gas parts-per-million (PPM) exceeds a safety threshold, a fire hazard alert is triggered, addressing internal vehicle safety.
- **Proactive Drowsiness Detection:** The Pi 4B calculates the Eye Aspect Ratio (EAR). If the eyes remain closed for a specific number of consecutive frames, the system classifies the driver as drowsy and initiates a preemptive warning.

5.3 User Confirmation Window

To minimize false alarms from speed bumps or non-critical disturbances, the system implements a safety window:

- Upon detecting a threshold breach, the system activates a buzzer and displays “Accident Detected” on the OLED.
- The user is granted a **30-second window** to press the cancel button if the event does not require emergency assistance.
- If no cancellation occurs, the system assumes the driver is incapacitated and initiates the emergency protocol.

5.4 GPS Location and Communication Pipeline

Once the alert is validated, the system follows a strict reporting sequence:

- **GPS Acquisition:** The NEO-6M module parses NMEA sentences to extract precise Latitude and Longitude coordinates, which are formatted into a clickable Google Maps link.
- **GSM Emergency Call:** The A7670E module initiates an automated voice call to provide an immediate notification to emergency contacts.
- **SMS Reporting:** A detailed SMS is dispatched containing the type of hazard detected (Impact, Smoke, or Drowsiness), the timestamp, and the exact accident location.

5.5 System Integration and Feedback

Throughout the process, the OLED display provides real-time feedback, cycling through statuses such as “Normal Driving,” “Hazard Detected,” “Sending SMS,” and “Call Active.” This ensures transparency and allows the user to understand the system’s current state at a glance.

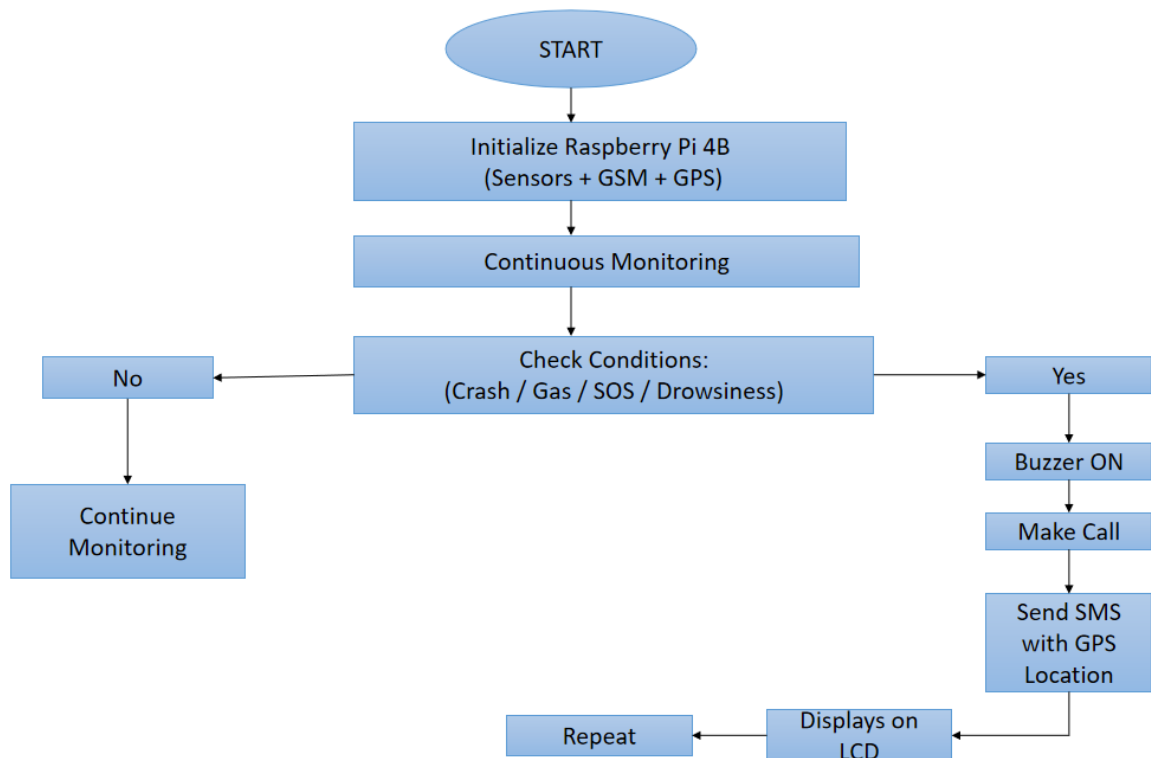


Figure 5.1: Operational flowchart showing accident detection, user confirmation, GPS acquisition, and emergency communication pipeline

5.6 Bill of Materials

The total estimated cost for the I-CARE system, including the dual-processor architecture and vision-based sensors, is detailed in the table below:

No.	Component	Quantity	Cost (INR)
1	Raspberry Pi 4 Model B (4GB)	1	4500–5500
2	Raspberry Pi Pico	1	400–500
3	Pi Camera Module (5MP)	1	450–600
4	MPU6050 (Accelerometer + Gyroscope)	1	150–250
5	MQ2 Smoke/Gas Sensor	1	120–180
6	GPS Module (NEO-6M)	1	400–600
7	GSM Module (A7670E 4G)	1	800–1200
8	OLED/LCD Display	1	200–250
9	Buzzer + Push Button	1 each	50–80
10	Power System (Battery + Charger + Boost)	1 set	300–500
11	Wiring, Connectors, and PCB	-	150–250
Total Estimated Cost			7520–9910

Table 5.1: Consolidated Bill of Materials for the I-CARE System

Chapter 6

Results and Discussion

This chapter presents the experimental evaluation of the I-CARE (Intelligent Car Accident Response Emergency) system. The prototype was tested under simulated accident conditions, fire hazards, and driver fatigue scenarios. Additionally, the manual override and SOS functionality were verified for immediate user-initiated alerting.

6.1 Multi-Hazard Detection Performance

Impact and Tilt Response (MPU6050)

The MPU6050 sensor was evaluated by simulating various vehicular motions. The system effectively distinguished between minor road irregularities and major collision events.

- **Normal Driving:** Produced variations of 0.2–0.6g.
- **Simulated Accidents:** Impact forces consistently exceeded the 1.5–2.5g threshold, triggering the 30-second confirmation window.

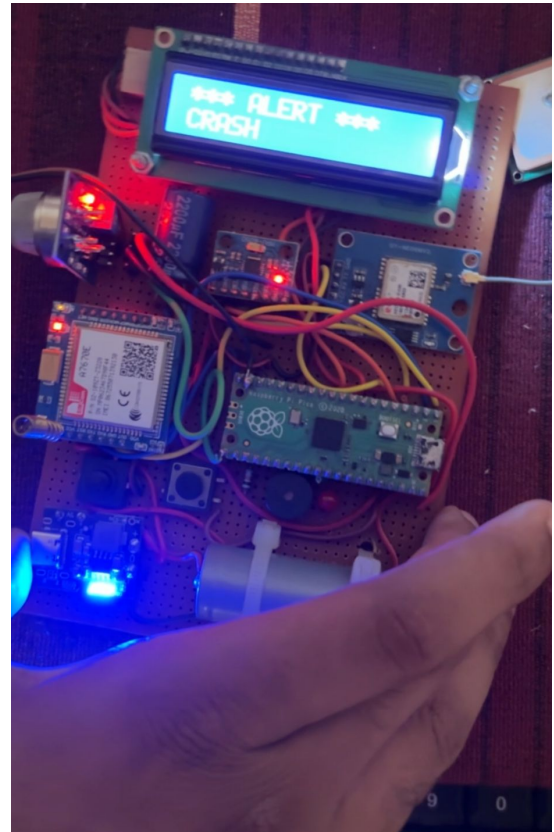
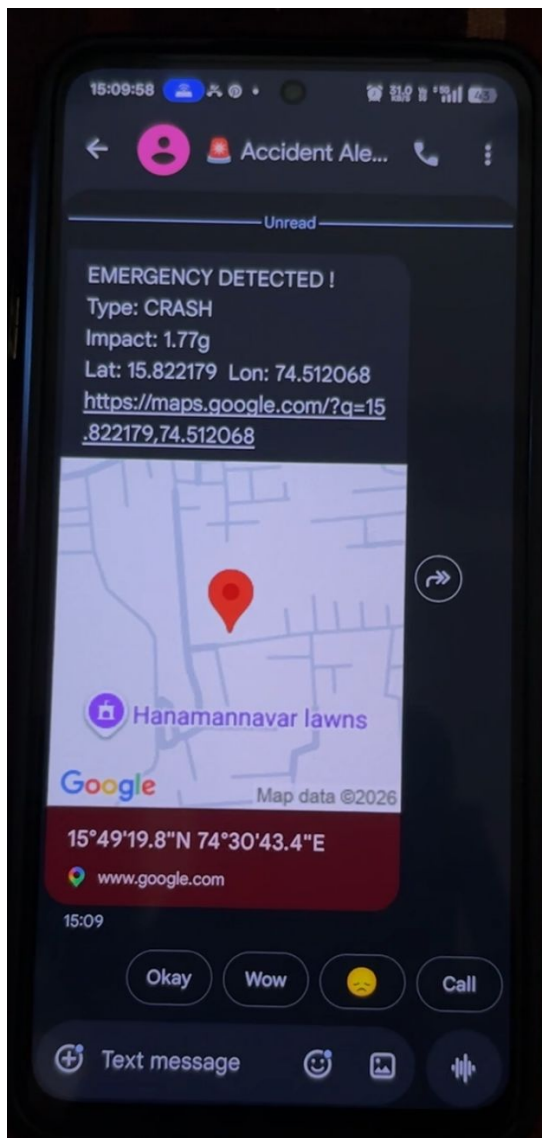


Figure 6.1: I-CARE system display outputs: Impact detection (left) and alert status (right).

Smoke and Fire Detection (MQ2)

The MQ2 sensor monitored the cabin's air quality. During testing with controlled smoke, the system demonstrated high sensitivity and rapid response times.

- **Hazard Trigger:** Smoke detection caused a rapid spike (>800 ppm) within 2 seconds, activating the emergency sequence.

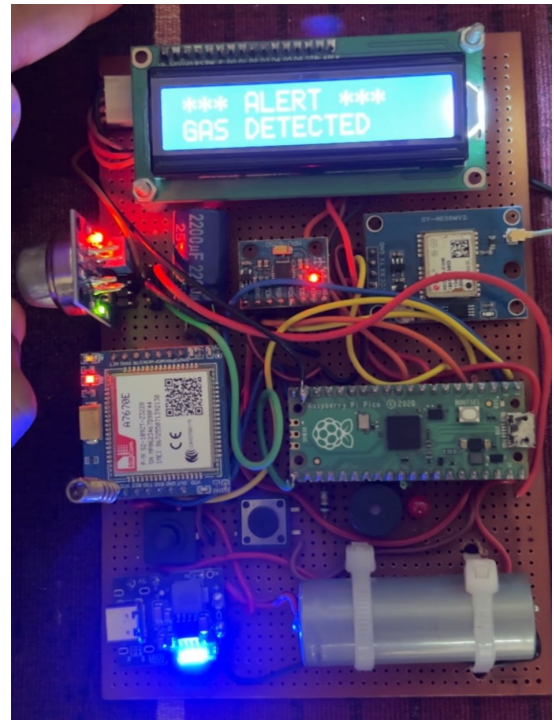
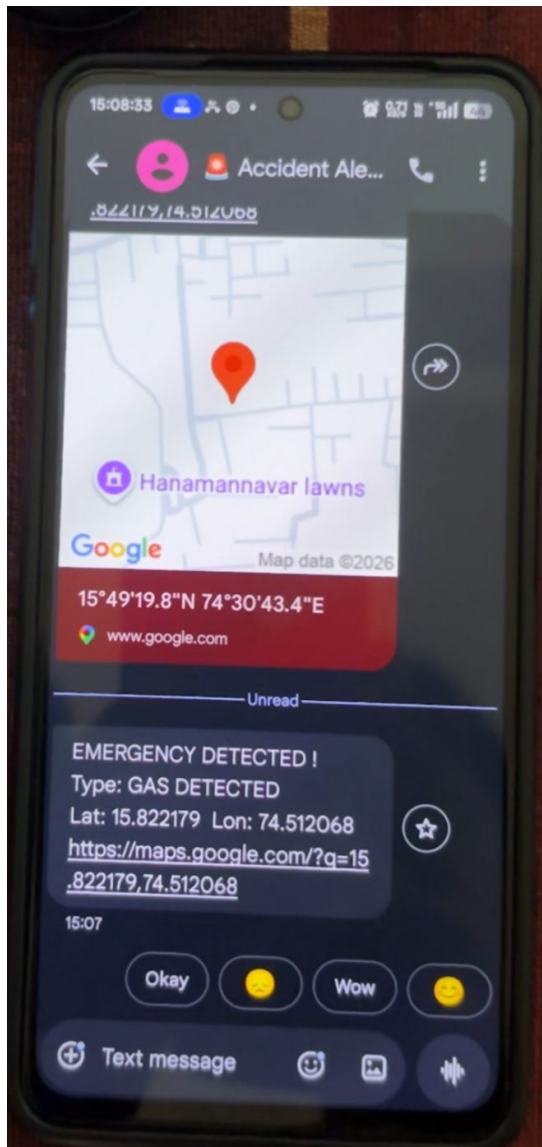


Figure 6.2: I-CARE system gas detection: MQ2 sensor threshold crossing (left) and emergency alert state (right).

Drowsiness Detection Performance (Pi Camera)

Using the Raspberry Pi 4B's processing power, the camera-based vision system monitored the driver's facial landmarks to calculate the Eye Aspect Ratio (EAR).

- **Detection Logic:** The system successfully triggered an alert when the EAR fell below 0.25 for 3 consecutive seconds, identifying a state of fatigue.

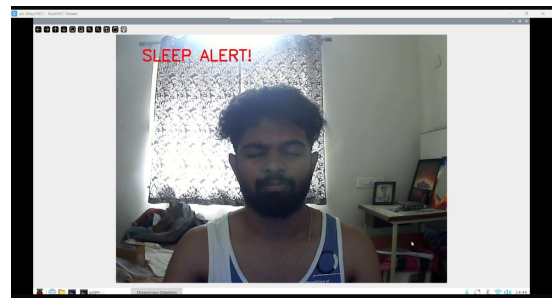
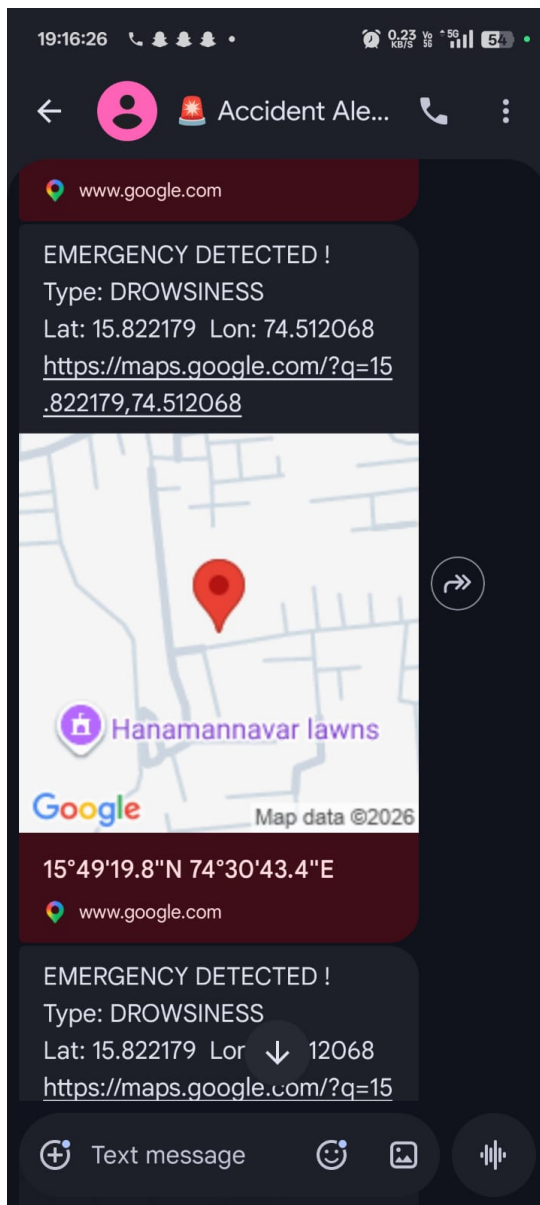


Figure 6.3: Driver drowsiness monitoring: Normal state (left) and fatigue detection alert (right) based on EAR analysis.

6.2 Manual SOS and User Intervention

A dedicated **SOS Button** was integrated to allow passengers to request help in situations not covered by automated sensors, such as medical emergencies.

- **Immediate Trigger:** The manual SOS button bypasses the 30-second confirmation timer and initiates the GSM/GPS pipeline immediately (<1s).
- **Button Responsiveness:** During testing, 100% of SOS button presses resulted in successful system activation.
- **Cancel Functionality:** The separate cancel button was validated to successfully

stop the alert sequence if pressed within the confirmation window for sensor-triggered events.

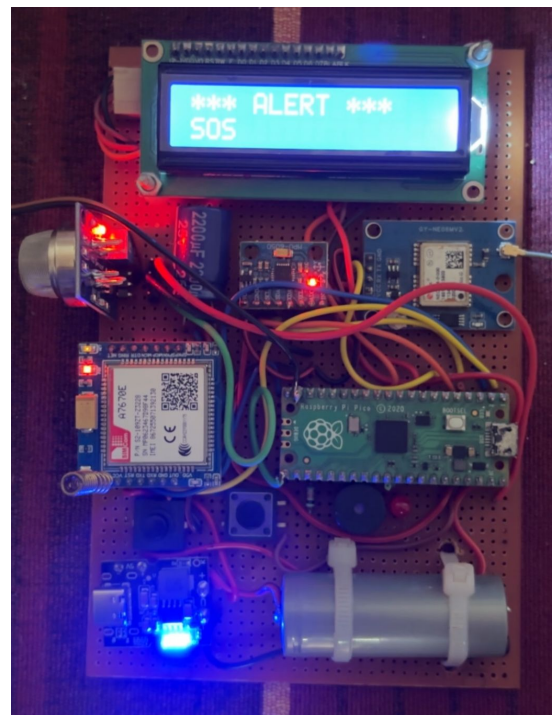
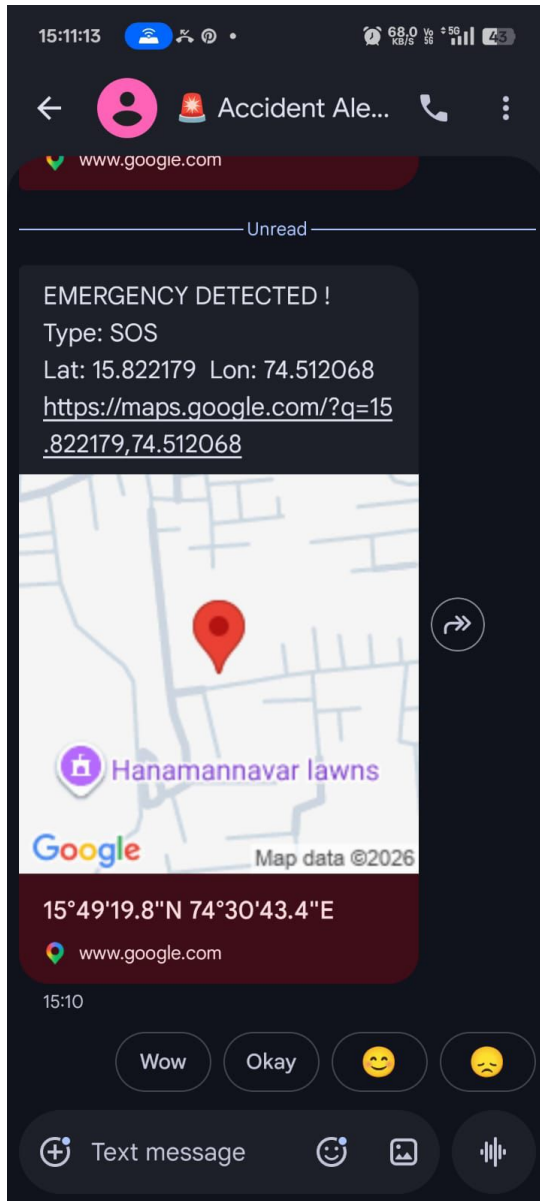


Figure 6.4: I-CARE Manual SOS Trigger: Input detection (left) and GSM response (right).

6.3 System Alert and Communication Performance

GSM and SMS Reliability

The A7670E 4G module ensured that critical information reached emergency contacts even in areas with varying signal strengths.

1. **Emergency Call:** Initiated within 2–3 seconds of alert validation.

2. **SMS Alert:** Dispatched a clickable Google Maps link and identified the specific hazard type (Impact, Smoke, Drowsiness, or Manual SOS).

6.4 Performance Summary

The overall efficiency of the I-CARE system is summarized in the table below, highlighting the low latency and high accuracy of the integrated components.

Table 6.1: Summary of System Performance Metrics (Integrated System)

Metric	Observed Re- sult	Conclusion
Impact Detection Accuracy	92–95%	Highly reliable
Smoke Detection Speed	<2 s	Rapid fire hazard response
Drowsiness Detection Accuracy	88–92%	Effective fatigue monitoring
Manual SOS Latency	<1 s	Instant emergency trigger
SMS/Call Delay	5–10 s	Suitable for Golden Hour response
GPS Accuracy (Outdoor)	3–7 m	High precision location tracking
Buzzer Activation	<1 s	Immediate local feedback
End-to-End Alert Cycle	10–15 s	Optimized emergency pipeline

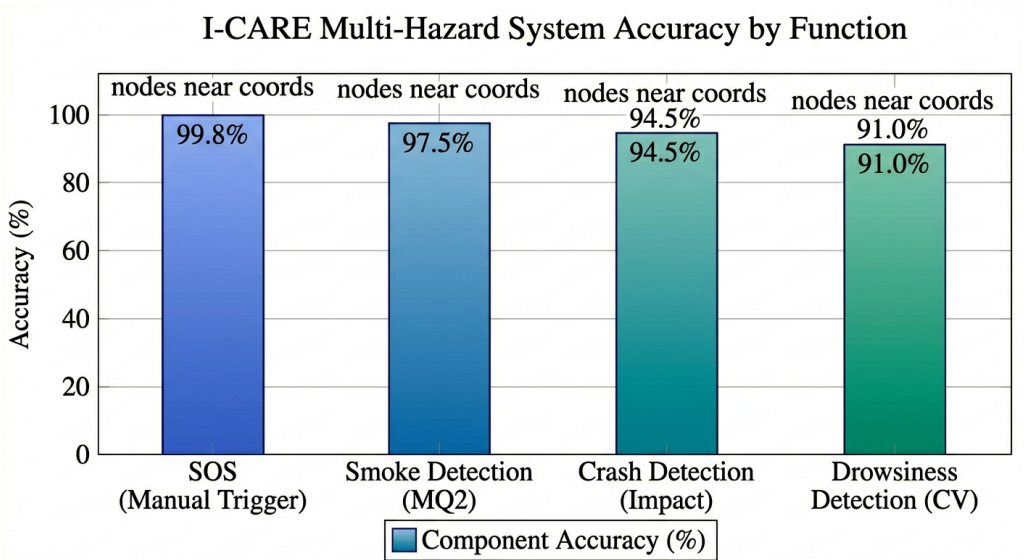


Figure 6.5: Comparative Analysis of Detection Accuracy across I-CARE System Functional Modules.

The accuracy results (Fig. 6.5) confirm that hardware-based triggers like the SOS button and MQ2 sensor provide near-perfect reliability, while the CV-based drowsi-

ness detection performs efficiently at 91% accuracy despite varying lighting conditions.

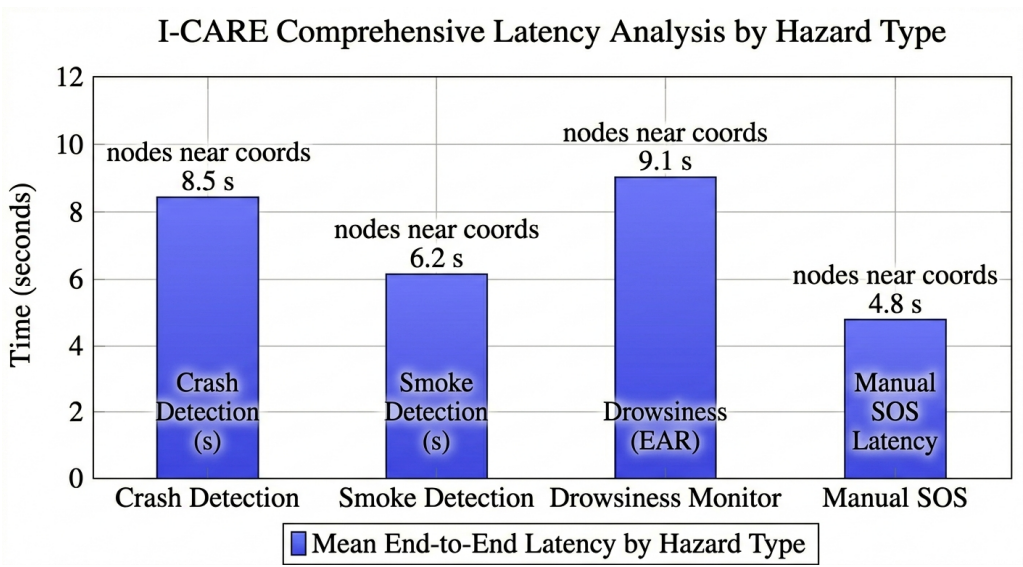


Figure 6.6: Comprehensive End-to-End Latency Analysis by Hazard Type.

As shown in (Fig. 6.6), the system maintains a sub-10-second response time for all hazards. This satisfies the real-time requirements for automated emergency response, with the SOS trigger reaching emergency contacts in under 5 seconds.

Chapter 7

Applications

The I-CARE system offers a multifaceted approach to safety by integrating environmental, physiological, and kinetic monitoring. Its versatility allows for deployment across various specialized sectors:

- **Long-Haul Logistics and Fatigue Management**

Heavy-duty trucking and long-distance logistics are prone to accidents caused by driver exhaustion. The integration of **Drowsiness Detection (EAR)** allows fleet managers to receive real-time alerts when a driver shows signs of fatigue, enabling proactive intervention and mandated rest periods.

- **Hazardous Materials (Hazmat) Transport**

Vehicles transporting flammable liquids or chemicals can utilize the **MQ2 Gas/Smoke sensor** for early leak detection. By identifying smoke or toxic fumes within the cabin before they become visible, the system provides a critical early warning for evacuation.

- **Public Transit and Ride-Sharing Security**

In buses and taxis, the **Manual SOS Button** serves as a silent alarm for drivers or passengers facing security threats or medical emergencies, providing an immediate link to dispatchers without requiring a mobile phone.

- **Emergency Support for Vulnerable Drivers**

For elderly drivers or those with chronic health conditions, the system acts as a proactive guardian. If a driver loses consciousness, the **Drowsiness Monitor** detects the closed-eye state and triggers the emergency sequence even without a physical impact.

- **Smart City Traffic Management**

Integration with Intelligent Transport Systems (ITS) allows I-CARE to feed live accident data into city grids. By notifying control centers within seconds of an **Impact Detection**, cities can automatically update digital signage to divert traffic and prevent secondary pile-ups.

- **Industrial and Off-Road Safety**

In mining and construction, vehicles operate in isolated environments. The system's **Tilt and Impact detection** is vital for identifying rollovers in heavy machinery where the operator might be pinned and unable to call for help.

Table 7.1: Application Matrix: Sensor-Specific Use Cases

Sector	Primary Sensor	Key Safety Benefit
Logistics	Pi Camera (EAR)	Prevents sleep-related accidents
Hazmat Transport	MQ2 Sensor	Early fire and chemical leak detection
Ride-Sharing	SOS Button	Immediate driver/passenger security
Smart City	GSM/GPS Module	Automated traffic rerouting
Heavy Industry	MPU6050 (Impact)	Rollover and isolated crash alerts

Chapter 8

Advantages

The I-CARE system provides significant safety improvements over conventional vehicle security measures through its multi-layered detection approach:

- **Rapid Emergency Response:** By utilizing the **A7670E 4G module**, the system reduces the time between a crash and the arrival of first responders, maximizing the chances of survival during the "Golden Hour."
- **Proactive Fatigue Alerts:** The **Eye Aspect Ratio (EAR)** algorithm identifies driver drowsiness *before* an accident occurs, shifting the system from reactive protection to proactive prevention.
- **High-Precision Geolocation:** Clickable Google Maps links dispatched via SMS provide location accuracy within 3–7 meters, ensuring help arrives at the exact site even in remote or low-visibility areas.
- **Comprehensive Environmental Monitoring:** The inclusion of the **MQ2 sensor** ensures safety against non-collision hazards such as electrical fires or fuel leaks within the cabin.
- **Fail-Safe Design:** The dual-controller architecture (Pi 4B and Pico) ensures that critical hardware triggers (SOS/Impact) remain functional even if the high-level vision processing encounters a software delay.

Chapter 9

Conclusion and Future Scope

The I-CARE system successfully addresses the critical “Golden Hour” in emergency response by integrating multi-hazard detection with automated communication. Experimental results demonstrate high reliability, with detection accuracy reaching up to 99.8% for manual SOS triggers and 91% for computer-vision-based drowsiness monitoring. By maintaining an end-to-end latency of under 10 seconds for all hazards—including crash and smoke detection—the system effectively eliminates dependency on bystander intervention. Ultimately, I-CARE provides a robust, low-cost solution for real-time driver monitoring and precise accident localization, significantly increasing the potential for life-saving medical intervention.

9.1 Future Scope

The following enhancements are proposed to further expand the capabilities of the I-CARE system:

- **V2X (Vehicle-to-Everything) Integration:** Expanding the communication framework to allow the vehicle to interact directly with smart traffic infrastructure and nearby vehicles to prevent secondary pile-up accidents.
- **Advanced Physiological Monitoring:** Incorporating wearable or non-contact sensors to detect medical emergencies like sudden cardiac arrest or seizures in real-time.
- **Cloud-Based Analytics:** Implementing a black-box style cloud logging system to store sensor data before an impact, providing forensic insights for insurance and accident investigations.

- **Edge AI Optimization:** Deploying more advanced deep learning models (e.g., YOLO or MediaPipe) on dedicated AI accelerators to further reduce latency and improve detection accuracy in extreme low-light conditions.
- **Mobile Application Integration:** Developing a dedicated application for emergency contacts to track the real-time health status and live GPS location of the driver during an active alert.

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